ANNALS OF ONCOLOGY DRIVING INNOVATION IN ONCOLOGY

SPECIAL ARTICLE

ESMO adaptation of Lines of Systemic Therapy (EnLiST): a consensus framework for standardising the designation of lines of therapy in solid tumours

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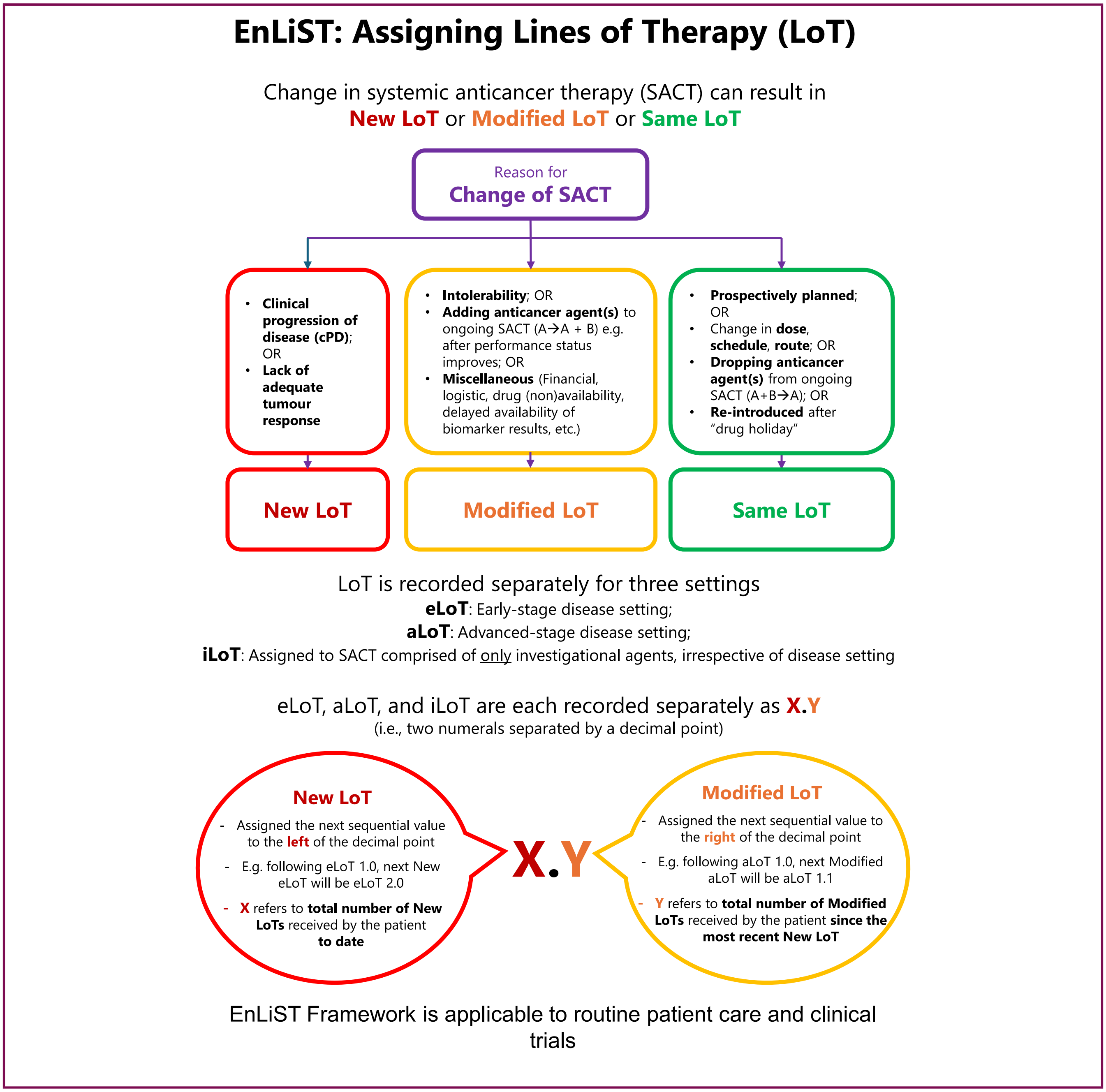
GRAPHICAL ABSTRACT



Enumeration of lines of therapy (LoT) is critical across oncology for ensuring optimal patient care, establishing uniform eligibility for clinical trial enrolment, standardising use of real-world data and pooling data for research purposes. Building on an earlier proposal for assigning LoT in oncology, the European Society for Medical Oncology (ESMO) developed the ESMO adaptation of Lines of Systemic Therapy (EnLiST) framework applicable to solid tumours across the full range of common clinical settings. A Delphi process was adopted to reach a consensus between expert representatives of multiple stakeholder perspectives including medical oncologists, clinical trialists, regulators, academics, patient advocates, experts from the pharmaceutical industry and clinical research organisations, artificial intelligence professionals, funding body specialists and ethicists. EnLiST provides a set of standard definitions, the format of reporting LoT, the minimum required data to be recorded and guidelines for assigning LoT. In EnLiST, a LoT is separately assigned to systemic anticancer therapies in the early (eLoT), advanced (aLoT) and investigational (iLoT) settings; each expressed as two numerals separated by a decimal point (e.g. 1.0). A change in systemic anticancer therapy results in either a 'New' LoT, a 'Modified' LoT, or the 'Same' LoT. A New LoT results from clinical progression of disease or lack of adequate tumour response, and is recorded as the next sequential value to the left of the decimal point (e.g. following LoT 1.0, the New LoT will be LoT 2.0). If the change in treatment is for other reasons (e.g. intolerability), a Modified LoT is recorded and assigned the next sequential value to the right of the decimal point (e.g. following LoT 1.0, the Modified LoT will be LoT 1.1). EnLiST has been designed to serve the needs of a wide range of stakeholders while enhancing the quality of clinical care, clinical research and real-world data.

Key words: lines of systemic therapy, systemic anticancer therapy, solid tumours, clinical trials, real-world data, consensus framework

INTRODUCTION

Cancer treatment and modalities have become increasingly complex.[1-4] The oncology literature is replete with inconsistent, confusing and sometimes conflicting use of terms relating to prior and ongoing lines of systemic anticancer therapy (SACT).[5-7]

For individualised, timely and informed treatment decisions, clinical trial eligibility, as well as for the interpretation of large datasets, a precise knowledge of the number, type and setting of prior systemic therapies is critical, given their impact on clinical outcomes, treatment efficacy and patient safety.[8,9] The accurate and uniform definition of the lines of therapy (LoT) is critically important for determining patient eligibility for clinical trials,[10-12] optimising their future treatment,[13,14] and adhering to approved drug labels and reimbursement policies.[15,16] Such standardisation would also positively impact on the conduct of clinical audits and health technology assessments in routine practice,[17,18] and the use of real-world data (RWD), for example in (national) cancer registries.[19,20] Furthermore, if linked to other datasets, such standardisation may provide important information regarding quality of life and patient-reported outcomes as well as facilitating the evaluation of prognostic factors, the comparative efficacy and the optimal sequencing of therapies.

Currently, in the absence of broadly accepted definitions and guidelines, there are many challenges involved in uniformly defining LoT. The simplest example is where a patient with a metastatic solid cancer has clearly documented radiological progression of their disease according to Response Evaluation Criteria in Solid Tumours (RECIST), and starts a new SACT regimen, and is assigned a new LoT. However, if the patient has symptomatic or biochemical progression as the basis for changing their systemic treatment, however, most treating oncologists may consider this to be a new LoT. Nonetheless, this would not have constituted progressive disease (PD) in the context of a clinical trial requiring central imaging review for the determination of radiological progression according to RECIST. Another example is where a patient does not have PD per RECIST criteria, but one component of a multi-agent SACT regimen is discontinued; this would likely not be considered a change in LoT. If a new drug is added to an ongoing SACT regimen in the absence of PD, however, this would be considered a new LoT by many oncologists. In some circumstances in the context of intolerability, one component of a regimen may be replaced by another with a similar, or different, mode of action; again, opinion would likely be divided as to whether this constitutes a new LoT or not. It is also not uncommon for patients with metastatic disease who are responding to a SACT regimen to be given a break from treatment (a 'treatment holiday'). Restarting the same regimen, however, may be characterised as a new LoT or as the same LoT depending on intervening therapies and disease progression or otherwise. The question of whether the enumeration of LoT should include SACT in both the early and metastatic settings is also a topic of debate.

While definitions and guidelines have been proposed for multiple myeloma,[21] to the best of our knowledge, no comprehensive guidelines exist for determining and reporting the LoT used for the treatment of patients with different solid cancers. The differences between the treatment of haematological malignancies and solid cancers are considerable, and the concept of early and metastatic disease settings are not typically applicable in the case of haematological malignancies. In the context of SACT in adult patients, NHS England offers guidelines on governance but does not define LoT or how to count it.[22] Similarly, while electronic prescribing platforms in widespread use invite the prescriber to assign each SACT a LoT, and often an intent or clinical context,[23] these systems do not provide standard definitions or guidelines for assigning LoT. A schema has been proposed in the context of insurance or reimbursement claims-based retrospective studies or based on RWD,[7,17] but they lack sufficient clinical context to be widely applicable. There is, therefore, a significant unmet need around standardising the enumeration of LoT for SACT in patients with solid tumours.

In 2021, Saini and Twelves[6] published the first systematic framework for determining LoT in patients with solid cancers. They highlighted the urgent need to develop and adopt a common vocabulary, and proposed uniform definitions, the recording of common minimum required data elements, and broad guidelines for counting LoT across the early- and advanced-stage disease settings. They also addressed the issue of what treatment modalities should constitute SACT in this context. These proposals were developed to be applicable to most of the common clinical scenarios encountered in the treatment of patients with solid malignancies, both in routine patient care and clinical trials.

The European Society for Medical Oncology (ESMO) was subsequently approached by Saini and Twelves and launched an initiative to adapt and further develop their framework.[6] ESMO set up and took the lead on the project which involved numerous ESMO experts to rework, complete, further develop and refine the initial framework proposed by Saini and Twelves[6] and to issue a new framework. The guiding principles of this new framework include its applicability across solid tumours and clinical settings, utility in both clinical practice and clinical trials and functionality regarding RWD and digital health studies. This included definition of the minimum required data for the assignment of LoT with minimal administrative burden to clinical teams, as well as ensuring clarity and consistency in the vocabulary used to define LoT. The result was creation of the new framework: **E**SMO adaptatio**n** of **Li**nes of **S**ystemic **T**herapy (EnLiST) Framework, which is presented in detail here for the first time.

METHODS

During the development of the EnLiST Framework, extensive consultation was conducted under the auspices of ESMO with experts representing a broad range of stakeholder perspectives including medical oncologists, clinical trialists, regulators, academics, patient advocates, experts from the pharmaceutical industry and clinical research organisations, RWD and digital health data analysts, artificial intelligence specialists, funding body specialists and ethicists. A chronological breakdown of the five phases and processes involved in the development of the EnLiST Framework is provided in Supplementary Table S1, available at https://doi.org/10.1016/j.annonc.2026.02.008.

Each element of the EnLiST Framework (including definitions of terms deemed key to establishing LoT, the format and the minimum required data for reporting LoT and the individual guidelines) was voted on using a Delphi process (phases 1 and 2).[24] Any elements that did not reach the predefined minimum 75% consensus underwent further revision and rounds of Delphi voting.

In order to evaluate the EnLiST Framework, and illustrate its utility, a series of tumour-specific clinical scenarios were established and assessed by experts in the following subgroups: breast cancer, lower digestive cancers, upper digestive cancers, gynaecological cancers, thoracic tumours, urologic cancers, sarcoma/adolescent and young adult tumours, head and neck/brain tumours, skin and miscellaneous cancers, clinical trial reporting and ethics as detailed in Supplementary Table S2, available at https://doi.org/10.1016/j.annonc.2026.02.008. These scenarios were used to develop vignettes for each of the EnLiST Framework guidelines, which were circulated to 547 members of the ESMO Faculty and members of relevant ESMO Working Groups in a survey regarding the acceptability of the EnLiST Framework recommendations (phase 3).

The results of the survey and all components of the framework were discussed at a hybrid face-to-face/virtual multistakeholder meeting (Brussels, 21 May 2024; phase 4) attended by 26 stakeholders (22 attended in person; 4 virtually). A final round of voting was undertaken to establish a consensus for all components of the finalised EnLiST Framework including the final guidelines for the enumeration of LoT (phase 5). Finally, minor text adjustments were made by the core team to ensure perfect clarity and intelligibility of the EnLiST framework, further validated through final manuscript revision by all authors.

A series of frequently asked questions regarding the EnLiST Framework along with the answers were generated (Supplementary Table S3, available at https://doi.org/10.1016/j.annonc.2026.02.008).

RESULTS

The EnLiST Framework was refined through multiple rounds of Delphi voting (for the statements and consensus for each round of voting, see Supplementary Tables S4-S9, available at https://doi.org/10.1016/j.annonc.2026.02.008), reaching a final consensus of $\geq$87.5% agreement for each element of the EnLiST Framework. For the final consensus for all elements, see Supplementary Table S10, available at https://doi.org/10.1016/j.annonc.2026.02.008.

THE ENLIST FRAMEWORK

The EnLiST Framework consists of four components: (i) definitions; (ii) format for reporting LoT; (iii) minimum required data for reporting LoT; and (iv) guidelines for uniform enumeration of LoT.

The development and refinement of each is discussed in detail below.

i Definitions

To minimise ambiguity in the vocabulary used in the assignment and counting of LoT, several key terms commonly used in the clinical setting needed to be defined. Definitions, including some originally proposed by Saini and Twelves,[6] were finalised and/or validated for the following final list of terms: (i) anticancer modality, (ii) clinical progression of disease (cPD), (iii) clinical setting (i.e. early-stage or advanced-stage disease), (iv) intolerability, (v) SACT, (vi) LoT and (vii) treatment intent. The definitions can be seen in Table 1 (all achieved $\geq$91.7% consensus after four rounds of voting; Supplementary Table S10, available at https://doi.org/10.1016/j.annonc.2026.02.008).

Most of these terms are intuitive, but others warrant attention. cPD can occur in the early- or advanced-stage disease setting, and is based on clinical signs, symptoms, laboratory results, with or without definitive radiological changes. Examples of trials that used laboratory results, instead of conventional RECIST radiological PD, to initiate new systemic therapy include SERENA-6 (NCT04964934), in which the detection of *ESR1* mutations in circulating tumour DNA was used to guide treatment changes ahead of conventional clinical or radiological PD,[25] and CHAARTED

Table 1. EnLiST definition of terms

Term	Definition	Notes/comments
1. Anticancer modality	Therapy used to remove, kill, or suppress cancer cells, including: (i) systemic anticancer therapy (SACT; see separate definition term 5 below); (ii) surgery; (iii) radiotherapy (including, but not limited to, external beam radiation, brachytherapy, proton beam, stereotactic radiosurgery, radioactive microspheres) but excluding radiopharmaceuticals (e.g. radioisotopes and radiolabelled monoclonal antibodies); (iv) other (including, but not limited to, high-intensity focused ultrasound, cryotherapy, thermal ablation, photodynamic therapy, hyperthermia, vascular embolisation, local administration of SACT regimen where there is no relevant systemic drug exposure in the opinion of the treating clinician, and anticancer agents not meeting the definition of SACT regimen).	Since radioisotopes such as sodium iodide (I-131), and radiolabelled monoclonal antibodies such as lutetium dotatate (Lu-177), have systemic effects, they are included in the definition of SACT, rather than external beam radiotherapy or brachytherapy, which have mainly locoregional effects.
2. Clinical progression of disease (cPD)	The clear worsening of the patient's clinical state due to malignancy, in the opinion of the treating clinician, OR the recurrence of malignancy, taking into consideration clinical findings, the appearance of or deterioration of clinical signs and/or symptoms, laboratory results, and/or imaging (including, but not limited to, objective imaging response criteria).	(i) Objective imaging response criteria include RECIST and/or other internationally accepted guidelines. (ii) The clinician should align cPD, where possible, with the definition of progressive disease by objective imaging response criteria. (iii) The diagnosis of cPD would usually necessitate consideration of a different anticancer therapy (systemic, locoregional or both) or best supportive care alone.
3. Clinical setting	The maximum extent of cancer spread to date, in the opinion of the treating clinician, and denoted as (i) early-stage disease setting (e.g. resectable disease if indicated); (ii) advanced-stage disease setting (e.g. typically unresectable locally advanced and/or distant metastases).	
4. Intolerability	Considerable toxicity in any organ system, and/or adverse physical, social, psychological, or emotional experience(s) occurring as a result of a SACT, that adversely impacts the patient's quality of life and leads to discontinuation of that SACT regimen.	
5. Systemic anticancer therapy (SACT)	A (bio)pharmaceutical product used for the systemic treatment of malignant disease, including—but not limited to—cytotoxic, endocrine, targeted, immunotherapy, cell and gene therapy and radiopharmaceuticals (e.g. radioisotopes and radiolabelled monoclonal antibodies) irrespective of route of administration, that has been approved by one or both of the EMA and United States FDA for treatment of any type or stage of cancer, and where the intent is to obtain a systemic (rather than a purely local/regional) effect. A SACT regimen consists of one or more systemic anticancer agents, which can be administered as single agents or in combination or sequence (which might include alternating, hybrid, continuation maintenance therapy [e.g. A + B→A] and/or switch to maintenance therapy [e.g. A + B→C]) Each SACT regimen is planned prospectively (i.e. when it starts) with any change in the agents pre-planned and not determined by lack of adequate response to treatment. It is usually (but not necessarily) administered in repeating cycles. It is administered systemically, or via local/regional routes but with the intention of a systemic effect or control of overall tumour burden; and is given at a clinically relevant dose for a duration that could be expected to exert a systemic anticancer effect.	(i) Types of systemic anticancer agents include: (i) cytotoxic therapy; (ii) endocrine therapy; (iii) targeted therapy; (iv) immunotherapy; (v) cell and gene therapy; (vi) anticancer vaccines; (vii) radiopharmaceuticals (e.g. radioisotopes and radiolabelled monoclonal antibodies); and (viii) other. (ii) A systemic anticancer agent in clinical development not yet approved by the EMA and/or the United States FDA for the treatment of any type or stage of cancer (as on the date of starting such therapy) is termed an 'investigational' systemic anticancer agent. (iii) SACT regimen agents may be 'Approved' or 'Investigational'. An approved SACT contains at least one anticancer agent approved by the EMA and/or the United States FDA for treatment of any type or stage of cancer (on the date of starting such therapy); it may also contain one or more investigational anticancer agent(s). An investigational SACT regimen is comprised exclusively of anticancer agent(s) that are investigational (on date of starting such therapy). (iv) A patient participating in a blinded randomised clinical trial where the control arm consists only of a placebo should not be considered to have received a SACT regimen unless unblinded information is available. (v) When a patient has two (or more) cancer diagnoses, a SACT regimen administered for one cancer should not be considered a SACT regimen for the other cancer(s), unless it is also considered by the treating clinician to be effective against the other cancer(s). (vi) An agent that is approved by the EMA and/or the United States FDA for a given cancer but used with the intention of providing only supportive/symptomatic care (but not direct anticancer effect) for another cancer (e.g. dexamethasone is approved for multiple myeloma, but may also be used as an antiemetic in patients with other cancers) should not be considered a SACT regimen for that other cancer.

Continued

Table 1. Continued

Term	Definition	Notes/comments
6. Line of therapy (LoT)	A serial chronological number assigned to each SACT regimen administered to a patient that denotes a discrete attempt to treat the cancer.	(i) A LoT falls into one of three categories: 1. Early LoT (eLoT) is assigned to a SACT regimen administered in the early-stage disease setting, usually with curative intent (e.g. resectable disease with no distant metastases); 2. Advanced LoT (aLoT) is assigned to a SACT regimen administered in the advanced-stage disease setting, usually with non-curative intent (e.g. unresectable locally advanced and/or distant metastases); and 3. Investigational LoT (iLoT) is assigned to a SACT regimen comprising only investigational agent(s) irrespective of the disease setting. (ii) LoT is separately assigned to eLoT, aLoT and iLoT, each expressed as Xx.Yy (i.e. two numerals separated by a decimal point). X refers to total number of New LoTs received by the patient to date. Y refers to total number of Modified LoTs received by the patient since the most recent New LoT (iii) If a change to an ongoing SACT occurs due to cPD or is prompted by lack of adequate tumour response, it results in a 'New LoT' which is assigned the next sequential value to the left of the decimal point (e.g. following LoT 1.0, the next New LoT will be LoT 2.0). (iv) By contrast, if the change in SACT is due to other reasons such as intolerability; OR adding anticancer agent(s) to ongoing SACT (A → A + B) e.g. after performance status improves; OR miscellaneous (e.g. financial, logistic, drug (non)availability, delayed availability of biomarker results, etc), it results in a 'Modified LoT' and is assigned the next sequential value to the right of the decimal point (e.g. following LoT 1.0, the next Modified LoT will be LoT 1.1). (v) Any change in a SACT regimen that is prospectively planned; OR results from a change in dose, schedule, route; OR dropping anticancer agent(s) from ongoing SACT (A + B → A); OR re-introduced after 'drug holiday' that is prospectively planned remains the same SACT regimen, and it results in the Same LoT being retained.
7. Treatment intent	Treatment intent, in the opinion of the treating clinician, can be curative or non-curative; the latter can be potentially life-extending or palliative: (i) Curative intent refers to therapy aimed at elimination of cancer and preventing its recurrence. (ii) Non-curative intent refers to therapy aimed at maintaining or improving the quality and/or the quantity of life (i.e. inducing 'remission') but without the expectation of cure.	

aLoT, line of therapy for the treatment of advanced-stage disease; cPD, clinical progression of disease; eLoT, line of therapy for the treatment of early-stage disease; EMA, European Medicines Agency; FDA, Food and Drug Administration; I-131, iodine-131; iLoT, investigational line of therapy; LoT, line of therapy; RECIST, Response Evaluation Criteria in Solid Tumours; SACT, systemic anticancer therapy.

(NCT00309985), in which a rising of prostate-specific antigen level ('PSA progression') was employed to confirm disease progression.[26]

SACT is an inclusive term encompassing therapies with different modes of action and routes of delivery that are linked by the intent to obtain a systemic (rather than a purely local/regional) effect. It is worth emphasising that a single SACT regimen may consist of one or more systemic anticancer agents, which can be administered as either a single agent, in combination, or in sequence, including maintenance therapy. Each SACT regimen is planned prospectively (i.e. before its initiation) with any change to the therapeutic agents pre-planned and not determined by a lack of adequate response to treatment.

ii Format for reporting lines of therapy

Numerical values for prior and current LoT are designated, for approved systemic therapies, according to treatment intent/setting of disease, or as an investigational therapy (91.7% consensus; Supplementary Table S10, available at https://doi.org/10.1016/j.annonc.2026.02.008). Three settings for a LoT have been defined:

- Early LoT (eLoT) is assigned to a SACT regimen administered in the early-stage disease setting (e.g. resectable disease if indicated)
- Advanced LoT (aLoT) is assigned to a SACT regimen administered in the advanced-stage disease setting (e.g. typically unresectable locally advanced and/or distant metastases)

- Investigational LoT (iLoT) is assigned to an investigational SACT regimen, irrespective of the early- or advanced-stage disease setting, if none of its components have been approved by the European Medicines Agency (EMA) and/or the United States Food and Drug Administration (FDA) for the treatment of any type or stage of cancer on the date of starting such therapy. See Table 1 for a definition of an investigational SACT regimen.

In modern oncology there is a great deal of complexity and uncertainty associated with the terms ‘curative’ and ‘non-curative’ or ‘palliative’ intent. Also, such characterisation may be a burden for patients and their families. Thus, we opted to use the terms eLoT and aLoT rather than curative and non-curative/palliative; (see Table 1, Definition 6). These replaced the notation of curative (cLoT) and palliative (pLoT) previously proposed by Saini and Twelves.[6]

The concept of treatment intent (i.e. curative versus non-curative disease) is intertwined with that of clinical setting (i.e. early versus advanced disease), and these terms are defined separately in Table 1. Locally advanced disease often involves increased tumour burden with invasion into adjacent structures or regional lymph nodes, typically unresectable or requiring multimodal treatment. We acknowledge that the optimal classification of an SACT regimen administered for locally advanced disease as eLoT or aLoT is a matter of debate, and context-dependent. For example, locally advanced testicular cancer would be expected to have a completely different clinical outcome than locally advanced pancreatic cancer. For most common epithelial cancers, locally advanced disease is generally associated with a higher risk of recurrence than early-stage disease, therefore aligning more closely with the clinical complexity of advanced-stage disease. Of note, the minimum required data for reporting LoT shown in Tables 2 and 3, contains recording of ‘Advanced (Locally advanced)’ disease separately from ‘Advanced (Metastatic)’ disease as well as the ‘curative’ or ‘non-curative’ intent, allowing future ‘road testing’ of the framework on this matter.

As outlined above, for any SACT regimen that consists entirely of agents that have not been approved by the EMA and/or the United States FDA for any type or stage of cancer (i.e. are only administered in the context of clinical trials or in programmes as investigational agents), the LoT is classified as investigational (iLoT). The rationale is that an iLoT is distinct from an eLoT or aLoT since its anticancer effect is not fully proven and hence should be captured separately.

The agreed format for reporting LoT includes early- and advanced-stage lines separately, but within parentheses, and any investigational lines outside the parentheses as shown below (100% consensus after three rounds of voting; Supplementary Table S10, available at https://doi.org/10.1016/j.annonc.2026.02.008):

[eLoT + aLoT] + iLoT

Importantly, this allows a distinction to be made between those LoT (i.e. eLoT and aLoT) that contain approved agents and those which do not (i.e. iLoT). Note that per definitions

Table 2. EnLiST minimum required data for reporting LoT

Minimum required data item	Subcategories/comments
1. Clinical setting	1a. Early 1bi. Advanced (locally advanced) 1bii. Advanced (metastatic)
2. Treatment intent	2a. Curative 2b. Non-curative
3. Date of most recent clinical progression of disease (cPD)	For patients newly diagnosed with cancer, the date of cPD is not applicable (NA); rather, the date of initial diagnosis should be used
4. Anticancer modality	4a: Systemic anticancer therapy (SACT) 4b: Investigational SACT 4c: Surgery 4d: Radiotherapy 4e: Other Note: 4a and 4b are each further subcategorised as: (i) cytotoxic therapy; (ii) endocrine therapy; (iii) targeted therapy; (iv) immunotherapy; (v) cell and gene therapy; (vi) anticancer vaccines; (vii) radiopharmaceuticals (e.g. radioisotopes and radiolabelled monoclonal antibodies); and (viii) other
5. Systemic anticancer agent(s)	—
6. Start date	Presented in YYYY-MM-DD format
7: Stop date	Presented in YYYY-MM-DD format
8: Reason for stopping	8a. cPD 8b. Completed planned anticancer therapy 8c. Intolerability 8d. Patient/clinician choice 8e. Death 8f. Other
9: Comments	—
10: Lines of therapy	[eLoT + aLoT] + iLoT

aLoT, line of therapy for the treatment of advanced-stage disease; cPD, clinical progression of disease; eLoT, line of therapy for the treatment of early-stage disease; iLoT, investigational line of therapy; NA, not applicable; SACT, systemic anticancer therapy.

presented in Table 1, an approved SACT contains at least one anticancer agent approved by the EMA and/or the United States FDA for treatment of any type or stage of cancer (on the date of starting such therapy); it may also contain one or more investigational anticancer agent(s). Table 3 gives an example of the use of this format of reporting LoT.

eLoT, aLoT and iLoT are each recorded separately as X.Y (i.e. two numerals separated by a decimal point). If a patient has not received one or more of the above, the value 0 should be recorded.

X refers to total number of New LoTs received by the patient to date; a New LoT is assigned the next sequential value to the left of the decimal point. Y refers to total number of Modified LoTs received by the patient since the most recent New LoT; a Modified LoT is assigned the next sequential value to the right of the decimal point. The criteria to consider a change in SACT as a New LoT or Modified LoT are described below in the guidelines for assigning LoT.

iii Minimum required data for reporting LoT

The agreed minimum required data for reporting LoT consists of 10 items (100% consensus after two rounds of voting; Supplementary Table S10, available at https://doi.

Table 3. Example of use of the minimum required data to determine the LoT in a patient with breast cancer

1. Clinical setting	2. Treatment intent	3. Date of most recent clinical progression of disease (cPD) YYYY-MM-DD	4. Anticancer modality	5. Systemic anticancer agent(s)	6. Start date YYYY-MM-DD	7. Stop date YYYY-MM-DD	8. Reason for stopping	9. Comments	10. Systemic lines of therapy: [eLoT + aLoT] + iLoT
1a Early	2a Curative	NA (date of initial diagnosis 1999-12-20)	4ai SACT cytotoxic; 4aii SACT endocrine; 4c Surgery; 4d radiotherapy	5FU, epirubicin, cyclophosphamide, tamoxifen, letrozole	2000-01-01	2002-08-14	8c Intolerability	Stage 3 breast cancer. FEC × 6 cycles → BCS → EBRT → Tam × 2 years → Let × 3 years planned (but patient developed intolerability)	[eLoT 1.0 + aLoT 0.0] + iLoT 0.0
1a Early	2a Curative	1999-12-20	4aii SACT endocrine	Anastrozole	2002-08-17	2005-09-10	8b Completed	Intolerability to letrozole. Changed to anastrozole	[eLoT 1.1 + aLoT 0.0] + iLoT 0.0
1a Early	2a Curative	2010-12-22	4aii SACT endocrine 4c Surgery	Exemestane	2011-01-01	2013-01-01	8b Completed	Isolated local recurrence. Resectable Mastectomy plus exemestane × 2 years	[eLoT 2.0 + aLoT 0.0] + iLoT 0.0
1bi Advanced (locally advanced)	2b Non-curative	2018-02-01	4aii SACT endocrine 4aiii SACT targeted	Ribociclib + fulvestrant	2018-02-07	2020-09-19	8a cPD	Non-resectable local recurrence	[eLoT 2.0 + aLoT 1.0] + iLoT 0.0
1bii Advanced (metastatic)	2b Non-curative	2020-09-19	4ai SACT cytotoxic	Paclitaxel	2020-09-27	2020-11-03	8f Other	*PIK3CA* test results became available (positive)	[eLoT 2.0 + aLoT 2.0] + iLoT 0.0
1bii Advanced (metastatic)	2b Non-curative	2020-09-19	4aii SACT endocrine 4aiii SACT targeted	Alpelisib + fulvestrant	2020-09-27	2021-11-03	8a cPD	Liver metastases	[eLoT 2.0 + aLoT 2.1] + iLoT 0.0
1bii Advanced (metastatic)	2b Non-curative	2021-12-01	4biii Investigational SACT targeted	First-in-human clinical trial with single-agent LAG3-targeting monoclonal antibody	2022-01-17	2022-03-05	8a cPD	Opted for hospice care after cPD	[eLoT 2.0 + aLoT 2.1] + iLoT 1.0

Treatment regimens mentioned are for illustrative purposes only, and do not necessarily represent standard of care.

Minimum required data subcategories: 1a: early; 1bi: advanced (locally advanced); 1bii: advanced (metastatic); 2a: curative; 2b: non-curative; 4a: systemic anticancer therapy SACT; 4b: investigational SACT; 4c: surgery; 4d: radiotherapy; 4e: other; note: 4a and 4b are each further subcategorised as (i) cytotoxic; (ii) endocrine; (iii) targeted; (iv) immunotherapy; (v) cell and gene therapy; (vi) anticancer vaccines; (vii) radiopharmaceuticals (e.g. radioisotopes and radiolabelled monoclonal antibodies); and (viii) other; 8a: cPD; 8b: completed planned anticancer therapy; 8c: intolerability; 8d: patient/clinician choice; 8e: death; 8f: other.

aLoT, line of therapy for the treatment of advanced-stage disease; BCS, breast-conserving surgery; cPD, clinical progression of disease; EBRT, electron beam radiotherapy; eLoT, line of therapy for the treatment of early-stage disease; FEC, 5-fluorouracil, epirubicin, cyclophosphamide; iLoT, Investigational line of therapy; Let, letrozole; NA, not applicable; SACT, systemic anticancer therapy; Tam, tamoxifen.

https://doi.org/10.1016/j.annonc.2026.02.008

org/10.1016/j.annonc.2026.02.008). Some of these items are sub-categorised further, as outlined in Table 2.

An example of how to capture the data to determine a LoT is presented in Table 3.

iv. Guidelines for assigning lines of therapy (with illustrative clinical vignettes)

Several versions of the guidelines originally proposed by Saini and Twelves[6] were revised and considered in an initial round of voting (Supplementary Table S6, available at https://doi.org/10.1016/j.annonc.2026.02), resulting in a total of seven guidelines following discussion and two further rounds of voting (consensus $\geq$83.3%; Supplementary Table S8, available at https://doi.org/10.1016/j.annonc.2026.02) . These guidelines, together with a series of clinical vignettes developed for illustrative purposes and the agreed set of definitions (Supplementary Appendix A, available at https://doi.org/10.1016/j.annonc.2026.02.008), were circulated to 547 ESMO members of including the ESMO Faculty and relevant ESMO Working Groups (phase 3). In total, 138 responses from the surveyed ESMO Faculty and Working Group members (response rate of 25%) were received (Supplementary Appendix B, available at https://doi.org/10.1016/j.annonc.2026.02.008). Only one guideline (guideline 5 of those circulated) failed to reach the predefined minimum 75% consensus [101/138 (73.2%) agreed; 30/138 (21.7%) disagreed and 7/138 (5.1%) abstained]. When, however, abstentions were removed, approval achieved the predefined minimum consensus [101/131 (77.1%)].

At the Brussels workshop (phase 4), the guidelines were further refined, and some amalgamated, to provide a final set of four guidelines, which were approved and agreed by all of the 26 stakeholder experts (100% consensus). The final four EnLiST guidelines are:

Guideline 1: ‘When a new SACT regimen is prompted by (a) cPD, or (b) a lack of adequate response to treatment in the absence of cPD, the new SACT regimen should be assigned a New LoT’ (Figure 1).

This guideline is applicable to most common scenarios when SACT is changed by the treating oncologist based on the clinically perceived resistance of the disease to the treatment received, due to either actual cPD or a lack of adequate response (e.g. a different adjuvant SACT is administered to patients with residual disease after neoadjuvant systemic therapy). A lack of adequate tumour response refers to a situation when an SACT does not produce the anticipated or clinically meaningful reduction in disease burden, e.g. in the setting of neo-adjuvant therapy for early breast cancer where insufficient response to an SACT may lead to a change in treatment.

Guideline 2: “When a SACT regimen is continued, or reintroduced, after cPD either on treatment or following a ‘treatment holiday’ without an intervening SACT regimen it (a) retains the Same LoT if there is no change in treatment setting or (b), if there is a change in the treatment setting from early to advanced, then it is assigned aLoT 1.0” (Figure 1).

This guideline stipulates a rule for the enumeration of LoT in the scenario of continued SACT beyond progression. The same SACT is sometimes continued after cPD, along with the addition of local ablative therapies, for the management of oligoprogression. It is also established clinical practice to sometimes suspend effective SACT for a ‘treatment holiday’ and then resume the same SACT without the use of any other intervening therapy. In most clinical scenarios, this would result in the Same LoT [guideline 2(a)]. In the relatively rare situation where an SACT is being given in the early-stage disease setting and following cPD the same SACT is continued in the advanced-stage disease setting, the SACT is recorded first as an eLoT, and then also as aLoT 1.0 [guideline 2(b)] since it was given in both settings.

Guideline 3: ‘In the absence of cPD, and when not determined by a lack of adequate response to treatment, if one or more anti-cancer agent(s) is/are discontinued from an ongoing SACT regimen and replaced by one or more agent(s), it is assigned a Modified LoT’ (Figure 1).

This guideline codifies the common situation in which an unplanned change of SACT is introduced due to reasons other than cPD or resistance, such as intolerability. In this case, discontinuation of the previous SACT and its replacement by a new SACT would biologically expose the patient, and the disease, to a different SACT that would have counted as a New LoT if introduced following cPD; therefore, there needs to be a mechanism to capture this critical information. Accordingly, per EnLiST, a change in SACT for reasons such as intolerability will not count toward the number of New LoTs a patient has received; this change should instead be recorded as a Modified LoT.

Guideline 4: “In the absence of cPD, and when not determined by a lack of adequate response to treatment, if one or more anti-cancer agent(s) is/are (a) discontinued from an ongoing SACT regimen, or if the same SACT regimen is re-introduced after a ‘treatment holiday’, the Same LoT is retained or (b) added to an ongoing SACT regimen, a Modified LoT is assigned” (Figure 1).

This guideline accounts for other less common treatment changes that are not driven by a perceived resistance of the disease to treatment (i.e. there is no cPD) and do not fit into Guideline 3. Whenever these changes result from the addition of one or more anticancer agent(s), the assignation of a Modified LoT records exposure of the patient and the disease to the different agent(s). When some agents from a multi-agent SACT are dropped, the remaining SACT agents retain the Same LoT.

A final Delphi round of voting of 40 experts was held following the Brussels workshop (phase V), and a consensus of $\geq$87.5% was reached for all four guidelines (Supplementary Table S9, available at https://doi.org/10.1016/j.annonc.2026.02.008).

The four guidelines of the EnLiST Framework are summarised and depicted visually in Figure 1. Examples of enumeration of LoTs are presented in Boxes A and B.

Figure 1. The EnLiST guidelines.
aLoT, line of therapy for the treatment of advanced-stage disease; cPD, clinical progression of disease; eLoT, line of therapy for the treatment of early-stage disease; iLoT, line of therapy assigned to SACT comprised of only investigational agent(s), irrespective of the disease setting; LoT, line of therapy; SACT, systemic anticancer therapy

DISCUSSION

To the best of our knowledge, the EnLiST Framework is the first comprehensive consensus document consisting of core definitions, a standard format for reporting LoT, the minimum required data for reporting LoT and detailed guidelines for uniform enumeration of LoT applicable to all solid tumours. Through an extensive five-phase process involving consultation with international key opinion leaders and multidisciplinary experts and the use of Delphi methodology, a broad consensus has been reached on a systematic, comprehensive and uniform framework applicable to most common clinical scenarios across the range of solid malignancies.

This consultation also identified several areas of controversy, the consideration of which guided the final outcome, as well as identifying some limitations of the EnLiST Framework. Given the variability in and evolving treatment approaches for different tumour types,[5,7] it is not surprising that there were some differences of opinion. An overall consensus was achieved, however, and EnLiST validated across all oncology disciplines. A list of frequently asked questions regarding the EnLiST Framework along with the answers, are displayed in Supplementary Table S3, available at https://doi.org/10.1016/j.annonc.2026.02.008.

We have developed the EnLiST Framework only around systemic therapies, given the vast array of cytotoxic, hormonal, targeted, immune-modulating and cellular therapeutics currently available. At present, we do not propose the assignment of LoT to locoregional therapies, such as surgery or radiotherapy. We recognise the potential existence of an abscopal effect, where radiation or another type of local therapy leads to the shrinkage of 'untreated' tumours elsewhere in the body.[27,28] The precise biological mechanism(s), frequency and clinical significance of abscopal effects, however, remain to be clarified.[28] Furthermore, our definition of SACT specifies that it has the 'intention of a systemic effect on or control of overall tumour burden; and is given at a clinically relevant dose for a duration that could be expected to exert a systemic anticancer effect'. Therefore, concurrent chemoradiotherapy (CRT) may or may not be assigned an LoT depending on the SACT dose and judgement of systemic effect. Of note, if CRT is combined with SACT, as with FOLFIRINOX in the total neoadjuvant treatment of rectal cancer, the SACT should be assigned a LoT. Assigning LoT to locoregional therapies may be reconsidered in future versions of the EnLiST Framework, especially if the role of local ablative therapies for retaining the same SACT beyond oligoprogressive disease becomes pertinent.

Similarly, we currently limit the EnLiST Framework to patients with invasive cancers and do not propose the assignment of LoT to chemoprevention strategies, such as the use of endocrine therapy in women at high risk of being diagnosed with invasive breast cancer or to those with ductal carcinoma *in situ*.

Another issue was how clinicians from different subspecialities currently count LoT when a patient responding to an SACT has a 'treatment holiday' and, with the development of cPD, is re-challenged with the same SACT administered before progression and without any intervening therapy. For instance, in the case of gynaecological cancers, re-treatment with a platinum-based regimen is commonly counted as a New LoT, whereas many gastrointestinal oncologists would count re-challenge with an oxaliplatin-based regimen as the Same LoT. In the EnLiST Framework guidelines for the assignment of LoT, the latter approach has been adopted, as covered by Guideline 2. Importantly, in the re-challenge scenario the Same LoT would not be retained had there been an intervening different SACT. Any challenges in implementing this particular guideline may be alleviated by the collection of detailed drug information within the minimum required data for recording LoT, which is a central critical component of the of EnLiST Framework.

A common principle followed in the EnLiST Framework is that exposure to a new SACT triggers a new LoT if it follows cPD or a lack of adequate tumour response. If the change in SACT is for any other reason (e.g. intolerability) and is replaced with another agent that is 'similar' (in structure and/or mode of action and efficacy), we considered whether the LoT should be unchanged. The definition of that 'similarity' is challenging, although Goldstein and colleagues[29] have proposed a framework for such a definition. We agreed, therefore, that any change in SACT under these circumstances (i.e. not due to cPD or lack of adequate tumour response) triggers a Modified LoT. For continuation or rechallenge scenarios, although they may not be biologically equivalent with respect to potential SACT efficacy differences, if the same SACT is continued beyond cPD or reintroduced with no intervening SACT the Same LoT is retained except when the treatment setting changes (i.e. from early-stage to advanced-stage disease). We also decided not to specify a duration of time off treatment that might define re-treatment as a new LoT, concluding that it would be impossible to implement this universally or to defend any arbitrary cut-off scientifically.

The concept of an 'investigational' LoT is, we believe, important. Regulatory approvals for new agents are often in the context of a specific line of treatment; similarly, reimbursement is often linked to the LoT. Failure to identify an LoT as investigational, and its incorporation simply as another eLoT or aLoT could have the unintended consequence of limiting access to, or reimbursement for, future SACT. It is important that clinical investigators, the pharmaceutical industry, clinical research organisations, patient advocates and other stakeholders work closely with scientific societies, regulators and reimbursement bodies to avoid such unintended consequences. Adoption of the EnLiST Framework in clinical protocols, and subsequent regulatory approvals and reimbursement agreements would facilitate transparency, harmonisation and equity of access.

We maintain that the EnLiST Framework, based on pragmatic and simple components, including guidelines for assigning LoT, covers the vast majority of common and relevant systemic treatment scenarios. Adoption of the EnLiST Framework in clinical trial protocols would help standardise the definition of trial populations and support regulatory approvals and reimbursement. The EnLiST Framework should benefit routine practice by the optimisation of the treatment sequencing landscape, supporting audits and outcomes as a means towards service improvement.[17,18] Standardisation of LoT enumeration and the adoption of the minimal required data for recording

LoT will also facilitate patient transition to referral centres for specialised treatments, including cross-border care and participation in clinical trials. More widely, the EnLiST Framework could also support the analysis and interpretation of SACT databases linked to electronic health records (EHR) and prescribing platforms where reliable LoT data are crucial but currently scanty, unreliable, or absent. There are also implications for real-world evidence studies, which regulatory authorities increasingly acknowledge[30] but where the current absence of standardised definitions and terminology partially addressed in other initiatives[31] remains a major challenge. The minimum required data captured within the EnLiST Framework (e.g. date of most recent cPD or treatment intent—frequently absent in EHR), will also provide a means for the standardisation of primary data collection in the context of the European Health Data Space which will facilitate the generalisability and interpretation of different RWD cohorts including (national) cancer registries.[32]

Looking to the future, we are keen to ‘road test’ the EnLiST Framework in different health care settings and interact with all stakeholders to validate an evolving and dynamic framework. To facilitate this, we are developing electronic versions of the EnLiST documentation and are keen to engage with stakeholders.

CONCLUSIONS

A proposed framework (EnLiST) for the enumeration and reporting of LoT for patients with solid tumours has been developed. It is anticipated that by adopting the EnLiST Framework, greater uniformity for assigning LoT will be achieved across a range of clinical, research, digital health and audit contexts that will assist in determining eligibility for clinical trials, optimising treatment decisions and adherence to guidelines, as well as streamlining primary data collection and enhancing RWD generation and analysis.

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SH declares data safety monitoring board role from Aveo Oncology, BeiGene, BMS, CG Oncology, Janssen and Sanofi; institutional funding from ASCO; associate editor role from ASCO. ELR declares speaker's engagement role from AstraZeneca; advisory board/consultancy role from Bayer, Janssen, LEO Pharma, Pfizer, Pierre Fabre, Roche, Seattle Genetics and Servier; institutional funding from BMS and Servier; non-remunerated board member role from the European Association of Neuro-Oncology (EANO); secretary role from EORTC. OLS declares invited speaker's engagement from AstraZeneca, Clovis and Lilly; advisory board role from GSK, MSD and Novartis; research grant from AstraZeneca and BMS. NBL declares institutional research grant from AstraZeneca, Inivata/NeoGenomics, Janssen, Lilly, Novartis and Pfizer; travel funding to deliver independent CME lectures from AstraZeneca, Guardant Health, Janssen, MSD and Sanofi; non-remunerated Data Safety Monitoring Board (DSMB) member from Daiichi Sankyo and Mirati. SLon declares speaker's engagement from Amgen, BMS, GSK, Lily, Merck Serono, MSD, Pierre Fabre, Roche and Servier; advisory board role from Amgen, Astellas, AstraZeneca, Bayer, BeiGene, BMS, Daiichi Sankyo, GSK, Helion, Incyte, Lilly, Merck Serono, MSD, Nimbus Therapeutics, Rottapharm, Servier and Takeda; coordinating PI role from Amgen, AstraZeneca, Bayer, BMS, Lilly, Merck Serono and Roche; non-remunerated member of the board of directors from the GONO Foundation. SLor declares speaker's engagement from AstraZeneca, BMS, Lily, MSD and Servier; advisory board role from Astellas. DL declares speaker's engagement from AstraZeneca, Clovis Oncology, Genmab, GSK, Immunogen, MSD, Novocure, Oncoinvest and Seagen; consultancy/advisory board role from AstraZeneca, Clovis Oncology, Corcept, Daiichi Sankyo, Genmab, GSK, Immunogen, MSD, Novartis, Novocure, Oncoinvest, PharmaMar, Seagen and Sutro; institutional funding/contracted research from Alkermes, AstraZeneca, Clovis Oncology, Corcept, Genmab, GSK, Immunogen, Incyte, MSD, Novartis, PharmaMar, Roche and Seagen; coordinating PI role from Clovis Oncology, Genmab and MSD; travel grants from AstraZeneca, Menarini, Clovis Oncology and GSK; non-remunerated PI role from AstraZeneca, Corcept, Clovis Oncology, Genmab, GSK, Immunogen, Incyte, MSD, Novartis, PharmaMar, Roche and Seagen; non-remunerated member role from ENGOT, non-remunerated board of directors role from the Gynecologic Cancer InterGroup (GCIG); non-remunerated president role from Multicenter Italian Trials in Ovarian Cancer and Gynaecological Malignancies (MITO). GMi declares speaker's engagement from Accuray, Brainlab; advisory board/consultancy role from Pfizer, Servier, AstraZeneca and Novocure; non-remunerated board member role from EANO; treasurer role from EORTC. GMo declares speaker's engagement from Amgen, AstraZeneca, MSD, Novartis and Pfizer; advisory board role from BMS, Janssen, Roche, Sanofi and Takeda; local PI role from Amgen, AstraZeneca, BMS, Gilead Pharmaceuticals, GSK, Lilly, MSD, Novartis and Roche; trial chair role from AstraZeneca. JO declares advisory board role from Astellas, AstraZeneca, Bayer, BMS, Eisai, Ipsen, Janssen-Cilaq, MSD, Pfizer and Roche; local/coordinating PI role from BMS, MSD and Sanofi. SO declares speaker's engagement from Merck and Travel Congress Management B.V.; advisory board role from BMS and Genmab; institutional research grant from Celldex, Merck, Novartis and Pfizer; non-remunerated steering and safety monitoring committee role from ALX Oncology; product samples from Celldex, Novartis and Pfizer. PO declares speaker's engagement from Amgen, AstraZeneca, Bayer, BMS, Eisai, Fresenius Kabi, Imedex, MSD, Nordic Drugs/Group, Pierre Fabre, Roche and Servier; advisory board/expert testimony role from Amgen, Astellas, AstraZeneca, Bayer, BMS, Daiichi Sankyo, Eisai, Incyte, Jansen Pharmaceutical, Johnson & Johnson, Merck, MSD, Pierre Fabre, Roche, Sanofi and Takeda; consulting/expert testimony role from BMS and Takeda; local/coordinating PI role from AstraZeneca, Bayer, Eli Lilly, Incyte, Merck, MSD, Nordic Drugs/Group, Pfizer, Roche, Sanofi and Servier; institutional research grant from Amgen; non-remunerated advisory board role from Servier and Scandion Oncology; non-remunerated member role from the Colores Patient Advocacy Group and the Finnish GI Cancer Group; non-remunerated board member role from the Finnish Cancer Society; non-remunerated board and president role from the Nordic Liver Cancer Organization (NOLCA). JLPG declares speaker's engagement from Roche; advisory board role from Astellas and MSD; local/coordinating PI role from Amgen, Astellas, BMS, Roche and Seattle Genetics; personal funding from Roche; institutional research grant from Roche. AP declares speaker's engagement form Servier; advisory board role from Merck and Takeda; travel grant from Merck; invitation to ENETS from Ipsen; non-remunerated member role from Fondation Française de Cancérologie digestive, and Groupe des Tumerus Neuroendocrines. SPN declares advisory board role from Deciphera, Immunocore, Pierre Fabre and Replimune. RP declares advisory board role from Astex Therapeutics, Bayer, BMS, CV6 Therapeutics, Cybrexa, Ellipses, Genmab, Immunocore, Incyte, Medivir, Novartis, Onexo, Pierre Fabre, Sanofi Aventis; royalties from Clovis Oncology; honoraria as member of the independent data monitoring committee (IDMC) from Alligator Biosciences, AstraZeneca, GSK and SOTIO. CP declares speaker's engagement from Angelini, BMS, Ipsen and MSD; advisory board/steering committee role from AstraZeneca, BMS, Eisai, Exelixis, Genenta, Genmab, Merck and MSD; safety monitoring committee role from Genenta; non-remunerated advisory role from AtG, Biorek and Medendi; non-remunerated member of the kidney cancer guidelines committee for the Italian Association of Medical Oncology (AIOM). KP declares speaker fees and honoraria for consultancy and advisory board functions for AstraZeneca, Axiom, Daiichi Sankyo, Eli Lilly, Exact Sciences, Focus Patient, Gilead Sciences, Medimix, Medscape, Menarini/Stemline, MSD, Need Inc., Nordic Pharma, Novartis, Pfizer, Hoffmann-La Roche, Sanofi, Seagen; institutional funding from Novartis; non-remunerated leadership role from BMSO; non-remunerated steering committee role from EORTC; non-remunerated advisory role from the Commission of Personalized Medicine, Federal Government Belgium and the European Medicines Agency. IRC declares advisory board role from Adaptimmune, Agenus, Amgen, AstraZeneca, BMS, Clovis, Daiichi Sankyo, Deciphera, EQRX, Eisai, GSK, Immunogen, MacroGenics, Merck Sereno, Mersana, MSD, Novartis, Oxnea, PMV Pharma, Seagen and Sutro; institutional translational research support from BMS; non-remunerated president role from GINECO; non-remunerated PI role from AstraZeneca (PAOLA1 trial). JR declares speaker's engagement from Janssen, Merck and Takea; travel and accommodation from Ose Immunotherapeutics; non-remunerated PI role from AstraZeneca, and

MSD; non-remunerated secretary role of the Lung Cancer Group of EORTC. CHR declares speaker's engagement from BMS, Helsinn Healthcare, MSD and Pharmanovia; consultant role from Astellas; coordinating PI role from Helsinn Healthcare and Novo Nordisk Foundation. PR declares speaker's engagement from AstraZeneca, BMS, Merck, MSD, Novartis, Pierre Fabre and Sanofi; advisory board role from Blueprint Medicines, BMS, Merck, MSD, Philogen, Pierre Fabre and Sanofi; institutional research grant from BMS and Pfizer; non-remunerated board of directors role from the Polish Oncological Society and the Polish Society of Surgical Oncology. LLS declares personal financial interest as advisory board role for Amgen, Arvinas, AstraZeneca, Daiichi Sankyo, GSK, LTZ Therapeutics, Marengo, Medicenna, Merck, Pangea, Pfizer, Relay Therapeutics, Roche, Tubulis, Voronoi; personal financial interest as spouse is cofounder of Treadwell Therapeutics and has stocks/shares in Agios; institutional financial interest as local PI for AbbVie, Amgen, AstraZeneca, Bayer, Boehringer Ingelheim, BMS, Daiichi Sankyo, EMD Serono, Gilead, GSK, Incyte, Marengo, Merck, Novartis, Pfizer, Roche/Genentech; non-financial interest as advisory role for Break Through Cancer, Dana Farber Harvard Cancer Center and ICR. SS declares speaker's engagement from Gentili; advisory board role from Agenus, Bayer, Boehringer, Daiichi, Ikena, Ipsen, NEC OncoImmunity AS, Pharma Essentia, Regeneron and Servier; travel funding from PharmaMar; local/coordinating PI role from Advenchen, Bayer, Daiichi, Deciphera, Epizyme, GSK, Hutchison MediPharma International Inc., Inhibrx, Karyopharm, Novartis, PharmaMar and SpringWorks; coinvestigator role from Boehringer; institutional funding from Blueprint; non-remunerated advisory role from Chordoma Foundation, Desmoid Foundation, Epithelioid Hemangioendothelioma Foundation, Epithelioid Haemangiodeothelioma Rare Cancer Charity; non-remunerated leadership role from the Italian Sarcoma Group; non-remunerated member of board of directors role from the Connective Tissue Oncology Society. CS declares speaker's engagement from Astellas Pharma, BMS, Hoffmann-La Roche Ltd, Ipsen and Merck; advisory board role from BMS, Eisai; Hoffman-La Roche Ltd, Ipsen and MSD; institutional funding/research grant from Ipsen and Pfizer. KS declares advisory board role from AbbVie, BMS and Pierre Fabre; honoraria for safety review committee from Sairopa; institutional research grant from BMS, Novartis, Philips, Genmab and TigaTx. JT declares speaker's engagement from Amgen, Astellas, Incyte, Merck, MSD, Novartis, Pierre Fabre, Servier, Takeda, Taiho; advisory board/expert testimony role from Amgen, Astellas, BMS, Brenus Pharma, Boehringer, Johnson & Johnson, Merck, MSD, Natera, Novartis, Pfizer, Pierre Fabre, Rottapharm, Sanofi, Servier and Takeda; steering committee role from Novartis; non-remunerated leadership role from the ARCAD Foundation, and the Federation Francophone de Cancerologie Digestive (FFCD); non-remunerated steering committee role from Pfizer and Servier. DT declares chair and working group role from WHO; working party role from the European Medicines Agency. AVa declares coordinating PI role from AstraZeneca, BMS and Novartis; institutional research grant from MSD and Roche. HWMvL declares speaker's engagement from Astellas, AstraZeneca, BeiGene, Benecke, Daiichi Sankyo, JAAP, MEDtalks, Novartis, Springer and Travel Congress management BV; consultant/advisory board role from Auristone, Incyte, Merck, Myeloid and Servier; institutional research grant and/or medication supply from Amphera, Anocca, Astellas, AstraZeneca, BeiGene, Boehringer, Daiichi Sankyo, Dragonfly, MSD, Myeloid, ORCA and Servier; selection of articles for Framingham; board of directors role from the Netherlands Comprehensive Cancer Organisation (IKNL); non-remunerated leadership role from ESMO. AVo declares speaker's engagement from AstraZeneca, BMS, Eisai, Ipsen, MSD and Roche; steering committee/advisory board role from AbbVie, Amgen, AstraZeneca, BeiGene, Boehringer Ingelheim, Bohringer Mannheim, Eisai, Incyte, Ipsen, Janssen, MSD, Roche, Servier, Taiho and Tyra. AZ declares speaker's engagement from MSD, Pfizer, Roche, and Takeda; advisory board role from AstraZeneca; consulting role from Gilead; safety review committee member role from Beyond Air; local PI role from AbbVie, AstraZeneca, Gilead and MSD; institutional research grant from BMS; stocks/shares from Nixio. GP declares employment from ESMO; non-remunerated membership of ASCO, Hellenic Cooperative Oncology Group (HeCOG) and Hellenic Society of Medical Oncology. SD declares speaker's engagement from Gilead; advisory board/steering committee role from BMS, Elsan, Gilead, Novartis, Roche Genentech and Sanofi; non-remunerated project lead role from the French government-based Banque des Territoires/France 2030 research project; non-remunerated PI role from the European Commission (H2020 funding); non-remunerated board of directors role from the Société Française de Sénologie et Pathologie Mammaire (SFSPM). CT declares speaker's engagement from Eisai, Lilly, Novartis, Menarini, Daiichi Sankyo and Pfizer; advisory board role from AstraZeneca, Pfizer and Daiichi Sankyo; travel and expenses from MSD, Pfizer and Novartis; non-remunerated member role ASCO and ESMO; unpaid trustee roles from the UK Charity for Triple Negative Breast Cancer and Breast Cancer UK. SBW has declared no conflicts of interest.

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